

Preliminary Assignment

Laura Betev, Ondrej Kerata, Julia Koziel,

Shirin Camille Elaine Schlamann, Rui Zhu

February 2026

1 Introduction

This report presents preliminary observations and exploratory analysis based on the provided dataset. We will investigate how image processing can be used to segment and classify skin lesions.

1.1 Description of the Data

The dataset consists of images of moles located on different areas of the body. Each image was visually examined and classified according to two specific criteria:

1. Hair Presence

Hair was rated on a scale:

0 = no visible hair

1 = small amount of hair

2 = moderate hair

3 = a lot of hair

This creates an ordinal variable, allowing us to distinguish not only presence but also intensity of hair.

2. Marker Presence

Marker was recorded as:

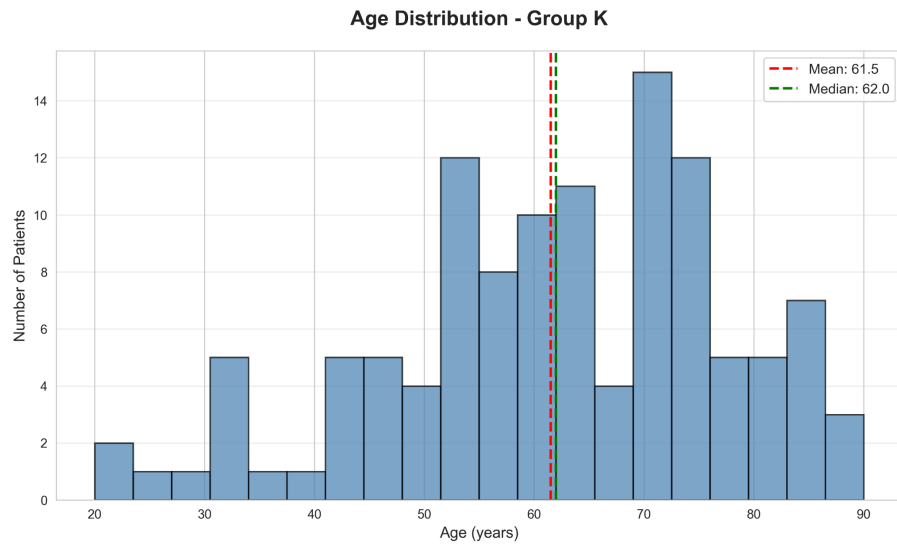
0 = no visible pen marking

1 = visible pen marking

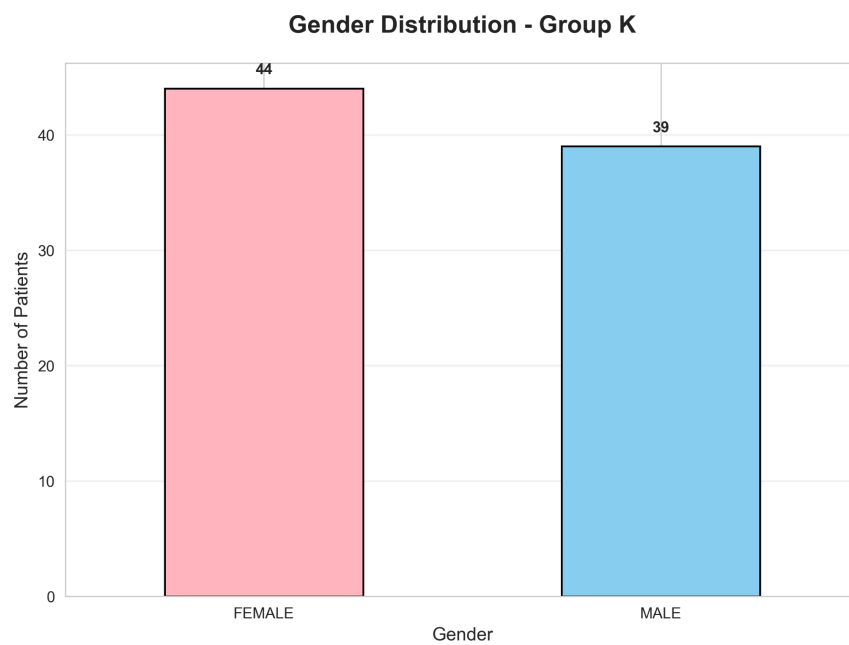
This variable is binary (categorical).

The classification was based on visual inspection of the images. No automated detection methods were used at this stage.

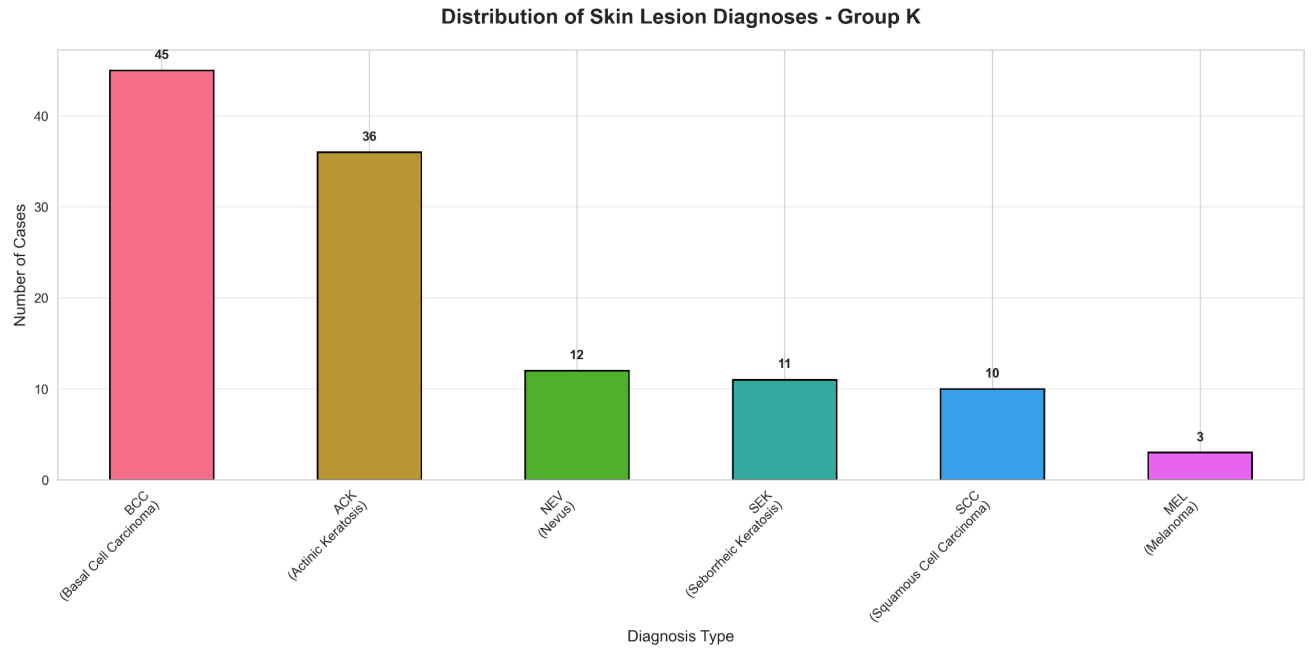
1.2 General Observations



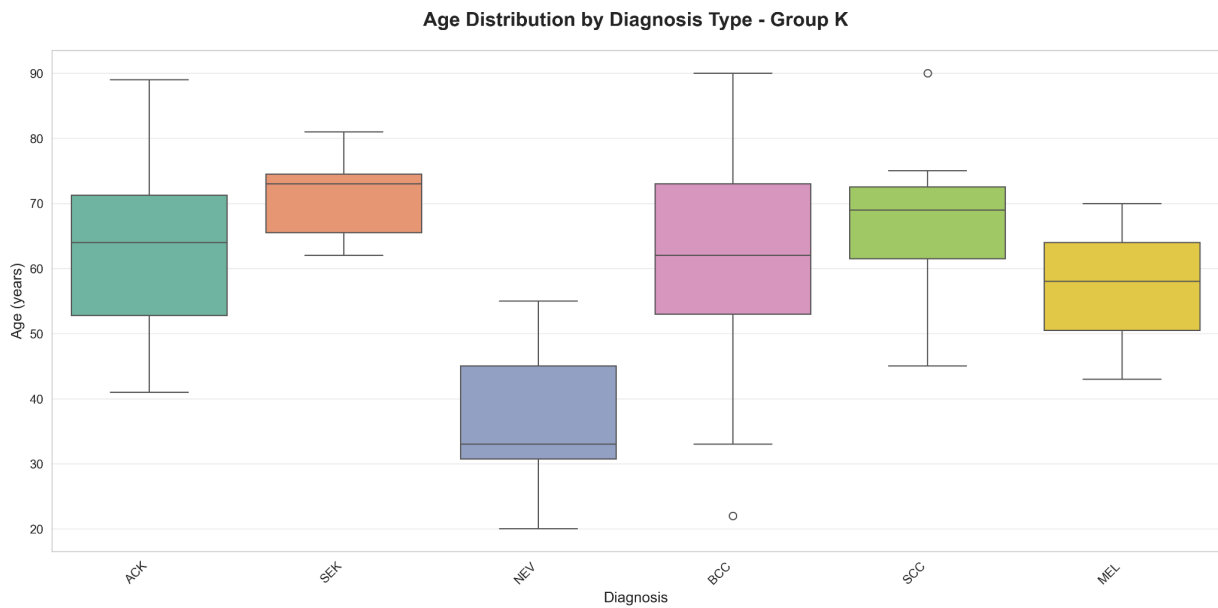
Distribution of age for group K



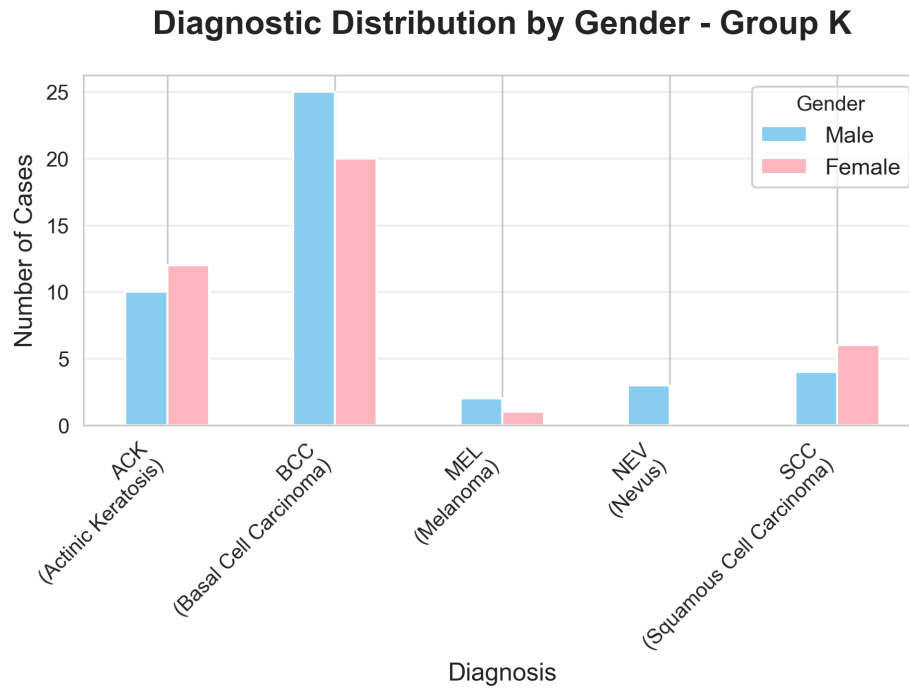
Distribution of Gender for Group K



Distribution of Types of Cancers



Distribution of Cancers by Age



Distribution of diagnosis by gender

Hair distribution

Preliminary inspection of the dataset suggests that:

A large proportion of images were rated as 0 (no hair) by most annotators.

A smaller subset of images consistently received high hair scores (2–3), indicating strong agreement in clearly hairy cases.

Moderate scores (1–2) appear less consistent across annotators, suggesting possible subjectivity in borderline cases.

This indicates that while extreme cases (no hair vs. a lot of hair) are relatively clear, intermediate cases may involve interpretation variability.

Marker distribution

The dataset is noticeably imbalanced toward “no marker.” The majority of the images were annotated as 0 (no marker), with marker 1 (no marker) being much less frequent.

Annotators are highly consistent when identifying markers. This suggests that the marker feature is relatively objective and clear to identify for annotators.

2 Inter-annotator agreement

Since the dataset was manually labeled, it is very important to evaluate the level of agreement between annotators.

2.1 Agreement on hair scores

This variable contains multiple ordered categories, and annotators can disagree on different levels of severity. For example, an annotator disagreeing between scores 0 and 1 is less significant than scores 0 and 3.

The extreme categories appear to have the strongest agreement, based on preliminary findings. Most annotators consistently classified images with no visible hair as 0. In a comparable way, pictures with a lot of visible hair typically scored a 3 with little dispute. These results suggest that annotators are likely to agree in their evaluations when visual evidence is clear and convincing.

On the other hand, middle categories (scoring 1 and 2) show more variability. Since there is no precise quantitative threshold that distinguishes "small amount of hair" from "moderate hair," these categories necessitate more subjective assessment. Inconsistent assessments could be caused by variations in how the density, thickness, or distribution of hair is perceived throughout

the lesion. It can be challenging to identify thin or light-colored hair in certain situations due to lighting or image quality issues.

2.2 Agreement on marker presence

Marker presence, as opposed to hair rating, was quantified as a binary variable (0 = no marker, 1 = visible marker). This classification significantly lowers the possibility of interpretative uncertainty because it just necessitates a yes-or-no choice.

For this variable, preliminary investigation shows that annotators have a high degree of agreement. With relatively few inconsistent annotations, the majority of photos were consistently classified as either having a visible pen marking or not. This implies that, in contrast to the more complex assessment of hair density, marker presence is more visually noticeable and simpler to spot.

2.3 Statistical analysis on the annotators agreement

We calculated agreement percentages for both variables (Hair and Marker) in order to quantitatively evaluate inter-annotator dependability. Three complementary measures were considered:

Full agreement: all five annotators assigned the same label.

Majority agreement: at least four out of five annotators agreed.

Average pairwise agreement: the average agreement across all possible annotator pairs.

These measures provide a more structured and objective evaluation of consistency than qualitative observation alone.

Hair (0–3):

The findings for the ordinal hair variable were as follows:

Full agreement (all 5 same): 45.30%

Majority agreement (≥ 4 agree): 75.21%

Average pairwise agreement: 71.71%

The gap between full agreement (45.30%) and majority agreement (75.21%) suggests that disagreement often occurs in borderline cases where one annotator assigns a slightly different intensity score (e.g., 1 instead of 2). The relatively high pairwise agreement (71.71%) indicates that, on average, annotators tend to agree, but not perfectly.

Overall, these results support the interpretation that hair classification is reasonably reliable but not entirely objective. The ordinal nature of the scale likely contributes to interpretative differences, especially when distinguishing between “small” and “moderate” hair presence.

MARKER (0–1):

For the binary marker variable, agreement was substantially higher:

Full agreement (all 5 same): 92.31%

Majority agreement (≥ 4 agree): 98.29%

Average pairwise agreement: 96.58%

From the data, it can be observed that agreement for hair was significantly lower than compared to marker presence. Full agreement for hair was less than half of the full agreement for marker presence, suggesting the difficulty of distinguishing the hair by the model in the future.

2.4 Correlation Analysis

To build on the agreement findings above, a correlation analysis was carried out to explore whether the annotation scores relate to the visual characteristics of the lesions. Hair consensus was calculated as the mean score across all five annotators, and marker consensus as the sum of positive annotations (0–5), where 5 means all annotators agreed a marker was present.

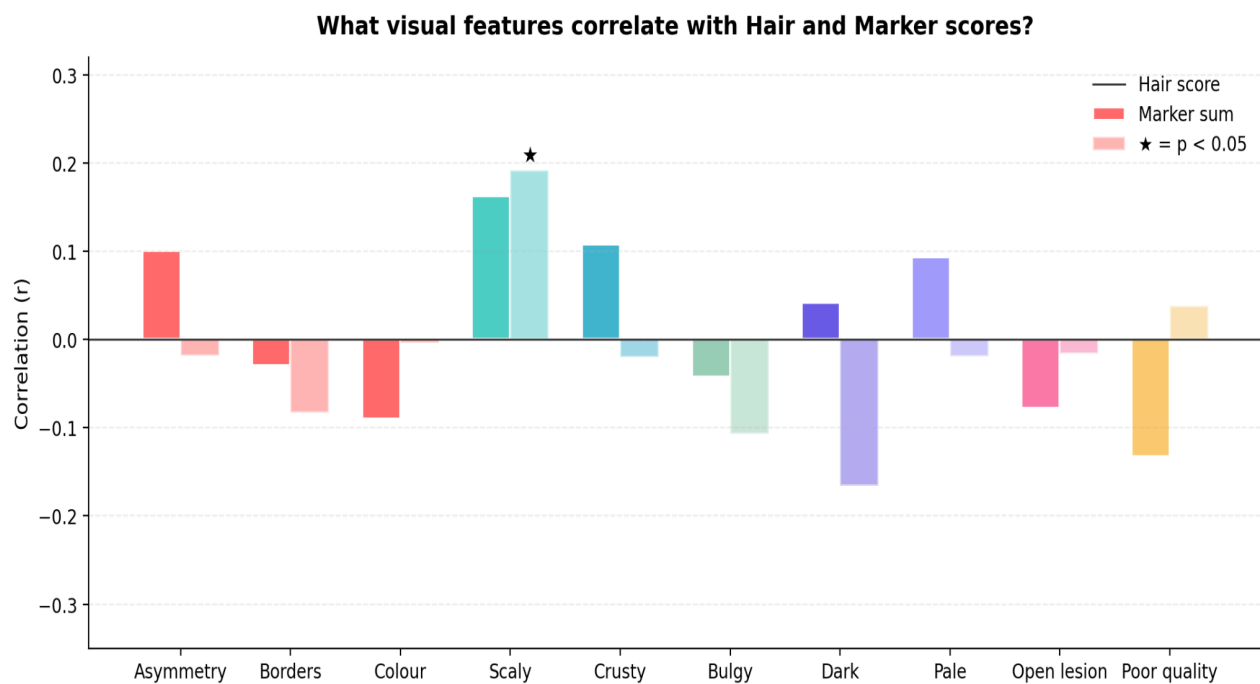
Hair and marker scores are not related to each other. The correlation between the two was essentially zero (Spearman $r = -0.04$, $p = 0.70$). This suggests that the presence of hair and the presence of a pen marker are capturing two completely different things in the image, and should be treated separately in future analysis.

Scaliness was the only visual feature that predicted marker placement. Ten binary visual descriptors were tested against both scores, including the ABCDE criteria, texture features, and pigmentation. Of all features, only scaliness showed a significant relationship with marker sum ($r = 0.19$, $p = 0.039$). A possible explanation is that scaly texture provides a clear visual cue that makes annotators more likely to identify and mark a lesion boundary. No feature significantly predicted hair scores.

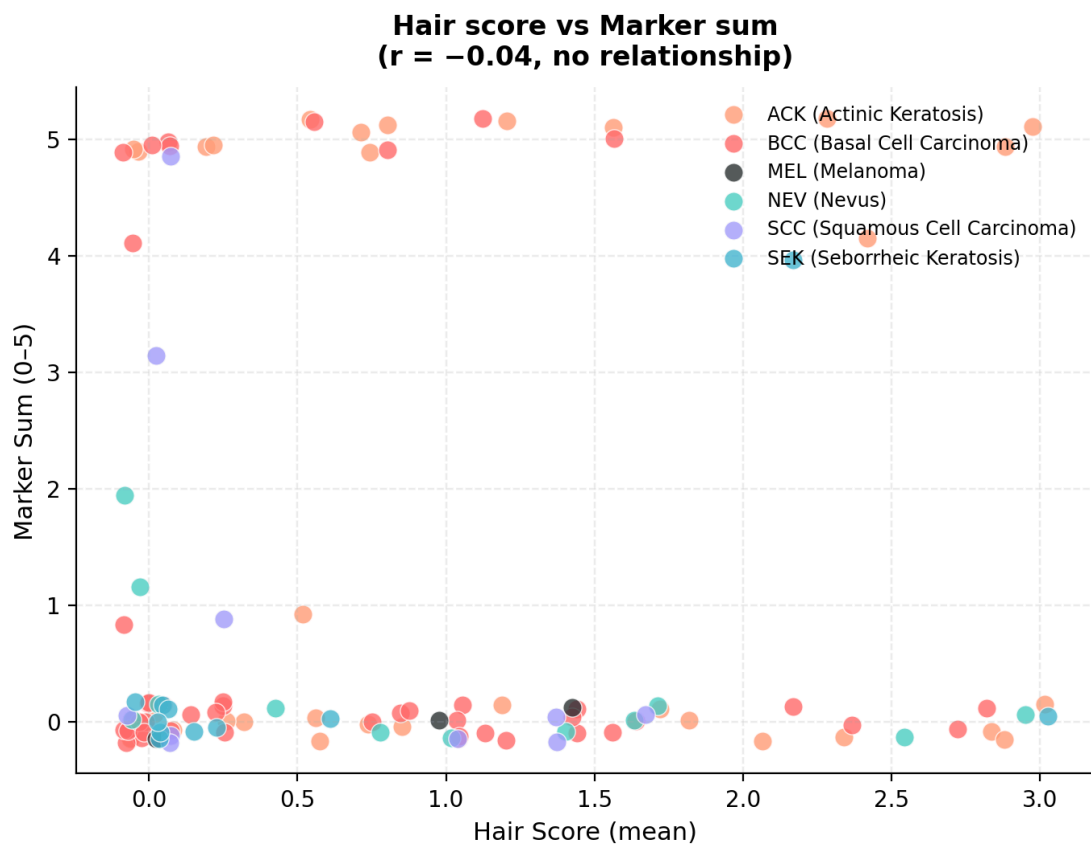
ABCDE criteria were unrelated to both scores. Asymmetry, irregular borders, and uneven colour showed no significant correlation with either hair or marker annotations (all $p > 0.27$). This is a

useful finding — it indicates that hair and marker labels are not simply reflecting clinical severity, but are genuinely independent dimensions of the image.

Annotator disagreement was not explained by any single feature. No visual descriptor was significantly associated with higher annotation variability (all $p > 0.08$), which supports the interpretation in Section 2.1 that disagreement in borderline cases arises from subtle perceptual differences rather than any clearly identifiable image property.



Correlation of ABCDE in our dataset



Correlation between hair score and marker score

3 Image quality and preprocessing challenges



Example of disagreement of hair annotation

ID: PAT_315_673_227

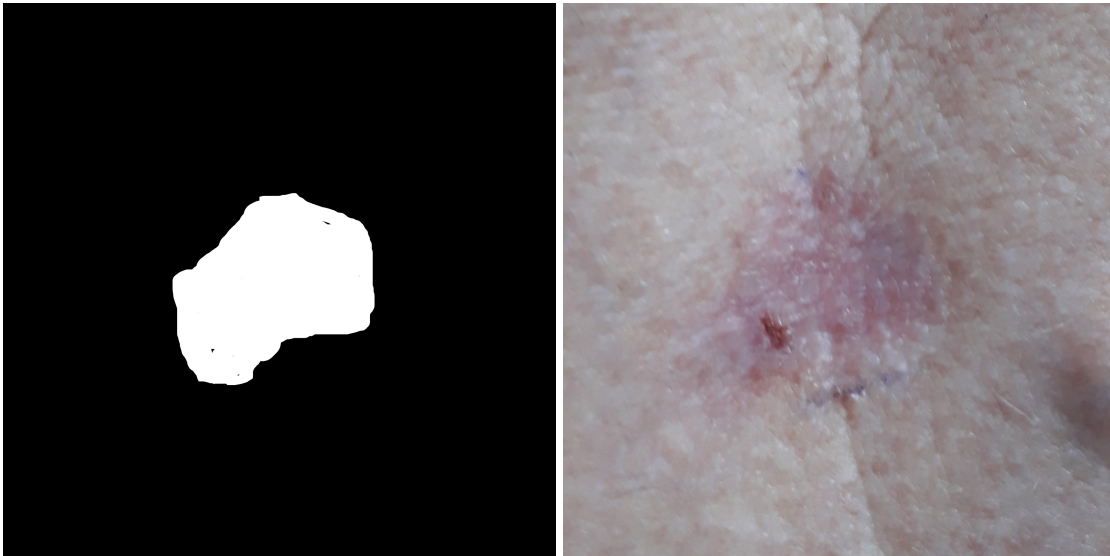
For the image above, most of the annotators had disagreements regarding the hair presence. There was a discrepancy between scores ranging from 0 to 2. This ambiguity may be because of the quality of the image but also the hair being so light.



Example of poor quality image and it's mask

ID: PAT_1061_254_813.png

The image above is a clear example of poor quality, the mask shows us the size of the mole but to the naked eye it clearly does not resemble the size. Making it very difficult to accurately distinguish the size of the mole.



Example of disagreement about pen markers

ID: PAT_847_1610_165.png

For the above image, annotators disagreed in respect to marker presence. Four out of five annotators marked the image with marker presence (1). This ambiguity could be due to poor image quality and the fact that the marker could be mistaken for discolouration.



Example of disagreement about pen markers

ID: PAT_2042_4301_636.png

For the above image, annotators disagreed in respect to marker presence. Two out of five annotators marked the image with marker presence (1). This ambiguity could be due to poor image quality and an unclear pen marking.

4 Conclusion

In this preliminary analysis, we explored the characteristics and annotation quality of a dataset of skin lesion images with respect to hair presence and marker presence. The results show that high inter-annotator agreement is caused by features that are visually distinct and apparent, such as excessive hair coverage, complete absence of hair, or obvious pen markings. However, there is more variation in intermediate hair classifications and ambiguous marker cases, showing the subjectivity of borderline visual evaluations. Practical issues with the dataset include discrepancies brought on by poor image quality and lighting.

Overall, this preliminary assessment shows the dataset's advantages and disadvantages. Hair classification adds subjectivity that needs to be carefully controlled in the analysis, while marker detection is precise. These results lay the essential groundwork for the project's next phase, which entails using and assessing image segmentation and classification methods. We can create more reliable methods in the future by comprehending the data properties and annotation reliability at this early stage.

Link to Group's Github:

https://github.com/shlockers/ProjectInDataScience2026_ExamTemplate